Grounded Explainability for Clinician End Users of a Conversational Medication Ranking System

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A medication ranking is hard to use in the clinic if the clinician cannot inspect it or argue with it. Many XAI papers still explain the model to the people who trained it. Clinicians ask different questions. They want to know why one option beat another, and they have to decide how far to follow a suggestion. Fluent text that sounds clinical can look like an explanation without tracking the model [1, 2]. A convincing explanation can also increase overreliance [3].

We are developing an explainable conversational medication recommender for clinical decision support. Conversational understanding turns a case description and structured record fields into a patient profile. A hybrid ranker then scores candidate medications. A Transformer branch models temporal and contextual dependencies in the stay. A graph neural network branch models relations among diagnoses, medications, and laboratory concepts. The two scores are fused into a ranked list for expert review. The software is a research prototype. Treatment decisions remain with the clinician. Historical ICU prescriptions from MIMIC-IV and eICU-CRD [4, 5] record what was given in past stays, not that a medication was the right choice.

The grounded explainability module is being developed as a separate layer from ranking, so the justification stays attached to evidence the ranker actually used. Current work on this module is about contrastive explanations of the fused ranking. For two candidates, it should show which patient factors moved the scores, which relations the graph branch used, whether encoded safety constraints agree with the ranking, and how uncertain that comparison is. The question we care about most is pairwise: why candidate A is above candidate B. Other drugs on the same list can change that comparison through slate normalisation, and that effect has to be visible. A language model may write the structured evidence in ordinary language. It must not add clinical claims that are not in the evidence. In the interaction we are working toward, the clinician can open an alternative and ask for a contrast. Each claim should show where it came from.

We are working on what the explanation contains and how it appears on screen. Evaluation belongs in that same work. We need to know if the text stays faithful to the fused ranker, and if clinicians can follow it. What it does to trust and reliance matters too. Raising trust on its own is a poor target. At the workshop we want to discuss how much computational detail belongs on a clinician's screen, and how to show a contrast so that it supports review. We also want to ask how reliance should be measured if agreeing with the model is not a good decision.

Keywords. explainable AI; clinician end users; grounded explanation; contrastive ranking; trust and reliance; clinical decision support.

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